

RETO 2 :CONFIGURACIÓN DE VLAN

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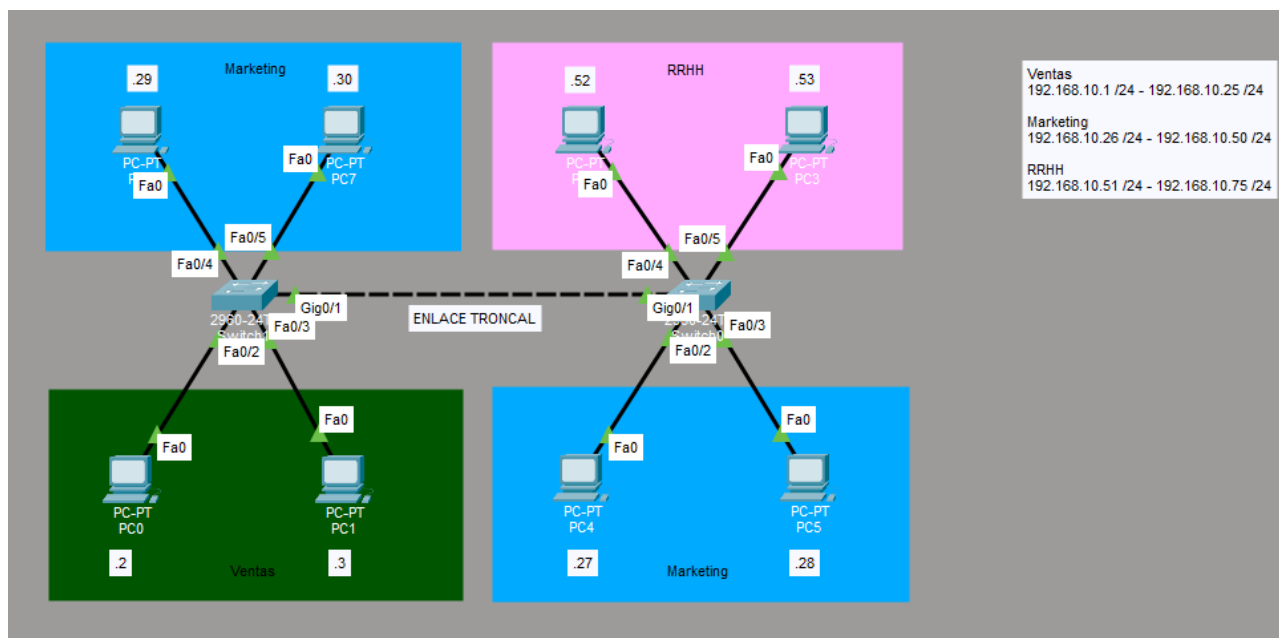
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1-Introducción

En este reto se diseña y configura una red local utilizando switches y VLAN mediante Cisco Packet Tracer. La red se divide en tres departamentos (Ventas, Marketing y Recursos Humanos) con el objetivo de separar el tráfico y mejorar la organización de la red.

Para ello se definen rangos de direcciones IP para cada departamento, se configuran las VLAN correspondientes en los switches y se realiza un enlace troncal entre ellos para permitir la comunicación entre equipos del mismo departamento situados en diferentes switches.

2-Diseño de la red



La red presenta una topología física en estrella extendida, diseñada en Cisco Packet Tracer mediante dos switches conectados entre sí a través de un enlace troncal. La red se divide en tres departamentos: Ventas, Marketing y Recursos Humanos, cada uno con dos equipos conectados a los switches.

Los ordenadores se conectan a los switches mediante cable de red directo (copper straight-through), ya que se trata de dispositivos diferentes. Tradicionalmente, las conexiones entre dispositivos del mismo tipo, como por ejemplo entre switches, se realizaban mediante cable cruzado.

El enlace entre switches se configura como un enlace troncal, lo que permite transportar las VLAN entre ambos dispositivos. De esta forma, cada departamento pertenece a una VLAN diferente y dispone de su propio rango de direcciones IP.

3-Planificación de direcciones IP

Breve explicación:

- Red utilizada: **192.168.10.0/24**
- Cada departamento utiliza su propio rango de direcciones IP.

Tabla de direcciones:

Ventas (VLAN 2)

Rango de IP: 192.168.10.1 – 192.168.10.25

Equipos:

- PC0 → 192.168.10.2
- PC1 → 192.168.10.3

Marketing (VLAN 3)

Rango de IP: 192.168.10.26 – 192.168.10.50

Equipos:

- PC2 → 192.168.10.29
- PC7 → 192.168.10.30
- PC4 → 192.168.10.27
- PC5 → 192.168.10.28

Recursos Humanos (VLAN 4)

Rango de IP: 192.168.10.51 – 192.168.10.75

Equipos:

- PC6 → 192.168.10.52
- PC3 → 192.168.10.53

Todos los equipos pertenecen a la misma red 192.168.10.0/24 para que la falta de conectividad entre departamentos se deba exclusivamente a la segmentación mediante VLAN.

4-Configuración de VLAN

4.1. Identificación de Switches

Se procede a cambiar el nombre de los switches para facilitar su identificación en la red.

Switch 1:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1-LUIS
SW1-LUIS(config)#end
SW1-LUIS#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
SW1-LUIS#exit
```

Switch 2:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2-LUIS
SW2-LUIS(config)#end
SW2-LUIS#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
SW2-LUIS#exit
```

4.2. Creación de VLAN y asignación de puertos

En este apartado se realiza la creación de las VLAN y la asignación de puertos en los switches según el departamento correspondiente.

Switch 1

```
SW1-LUIS>enable
SW1-LUIS#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1-LUIS(config)#vlan 3
SW1-LUIS(config-vlan)#name Marketing
SW1-LUIS(config-vlan)#exit
SW1-LUIS(config)#vlan 4
SW1-LUIS(config-vlan)#name RRHH
SW1-LUIS(config-vlan)#exit
SW1-LUIS(config)#interface range fa0/2-3
SW1-LUIS(config-if-range)#switchport access vlan 3
SW1-LUIS(config-if-range)#exit
SW1-LUIS(config)#interface range fa0/4-5
SW1-LUIS(config-if-range)#switchport access vlan 4
SW1-LUIS(config-if-range)#exit
SW1-LUIS(config)#end
SW1-LUIS#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

```
SW1-LUIS#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
3	Marketing	active	Fa0/2, Fa0/3
4	RRHH	active	Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1-LUIS#
```

Switch 2

```
SW2-LUIS>enable
SW2-LUIS#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW2-LUIS(config)#vlan 2
SW2-LUIS(config-vlan)#name Ventas
SW2-LUIS(config-vlan)#exit
SW2-LUIS(config)#vlan 3
SW2-LUIS(config-vlan)#name Marketing
SW2-LUIS(config-vlan)#exit
SW2-LUIS(config)#interface range fa0/2-3
SW2-LUIS(config-if-range)#switchport access vlan 2
SW2-LUIS(config-if-range)#exit
SW2-LUIS(config)#interface range fa0/4-5
SW2-LUIS(config-if-range)#switchport access vlan 3
SW2-LUIS(config-if-range)#exit
SW2-LUIS(config)#end
SW2-LUIS#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

```
SW2-LUIS#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
2	Ventas	active	Fa0/2, Fa0/3
3	Marketing	active	Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW2-LUIS#
```

5- Configuración del enlace troncal

Se ha configurado un enlace troncal entre los switches para permitir el paso de varias VLAN por un único enlace físico, permitiendo la comunicación entre equipos del mismo departamento en distintos switches, limitando las VLAN necesarias mediante **switchport trunk allowed vlan** para optimizar el tráfico y evitar información innecesaria.

Switch 1

```
SW1-LUIS>enable
SW1-LUIS#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW1-LUIS(config)#interface g0/1
SW1-LUIS(config-if)#switchport mode trunk

SW1-LUIS(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
switchport trunk native vlan 1
SW1-LUIS(config-if)#switchport trunk allowed vlan 1,3,4
SW1-LUIS(config-if)#end
SW1-LUIS#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
SW1-LUIS#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1,3-4

Port      Vlans allowed and active in management domain
Gig0/1    1,3,4

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    none
```

Switch 2

```
SW2-LUIS>enable
SW2-LUIS#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW2-LUIS(config)#interface g0/1
SW2-LUIS(config-if)#switchport mode trunk
SW2-LUIS(config-if)#switchport trunk native vlan 1
SW2-LUIS(config-if)#switchport trunk allowed vlan 1,2,3
SW2-LUIS(config-if)#end
SW2-LUIS#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
SW2-LUIS#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-3

Port      Vlans allowed and active in management domain
Gig0/1    1,2,3

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    3
```

6- Pruebas de conectividad

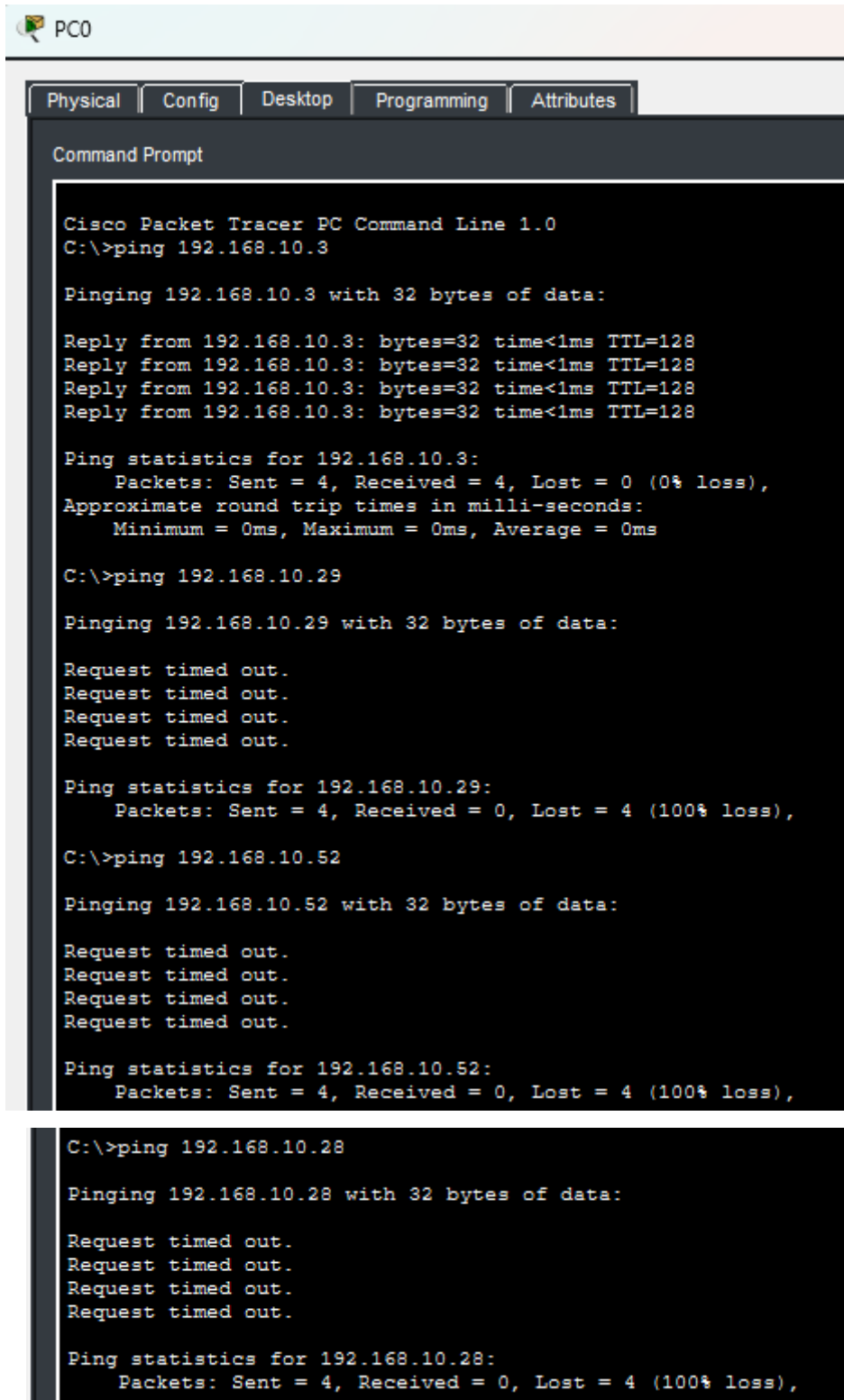
1-Desde PC0 (192.168.10.2) — Ventas (VLAN 2)

ping 192.168.10.3 - Ventas (OK)

ping 192.168.10.29 - Marketing switch 2 (FAIL)

ping 192.168.10.52 - RRHH (FAIL)

ping 192.168.10.28 – Marketing switch 1 (FAIL)



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.29

Pinging 192.168.10.29 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.29:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.52

Pinging 192.168.10.52 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.52:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.28

Pinging 192.168.10.28 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.28:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

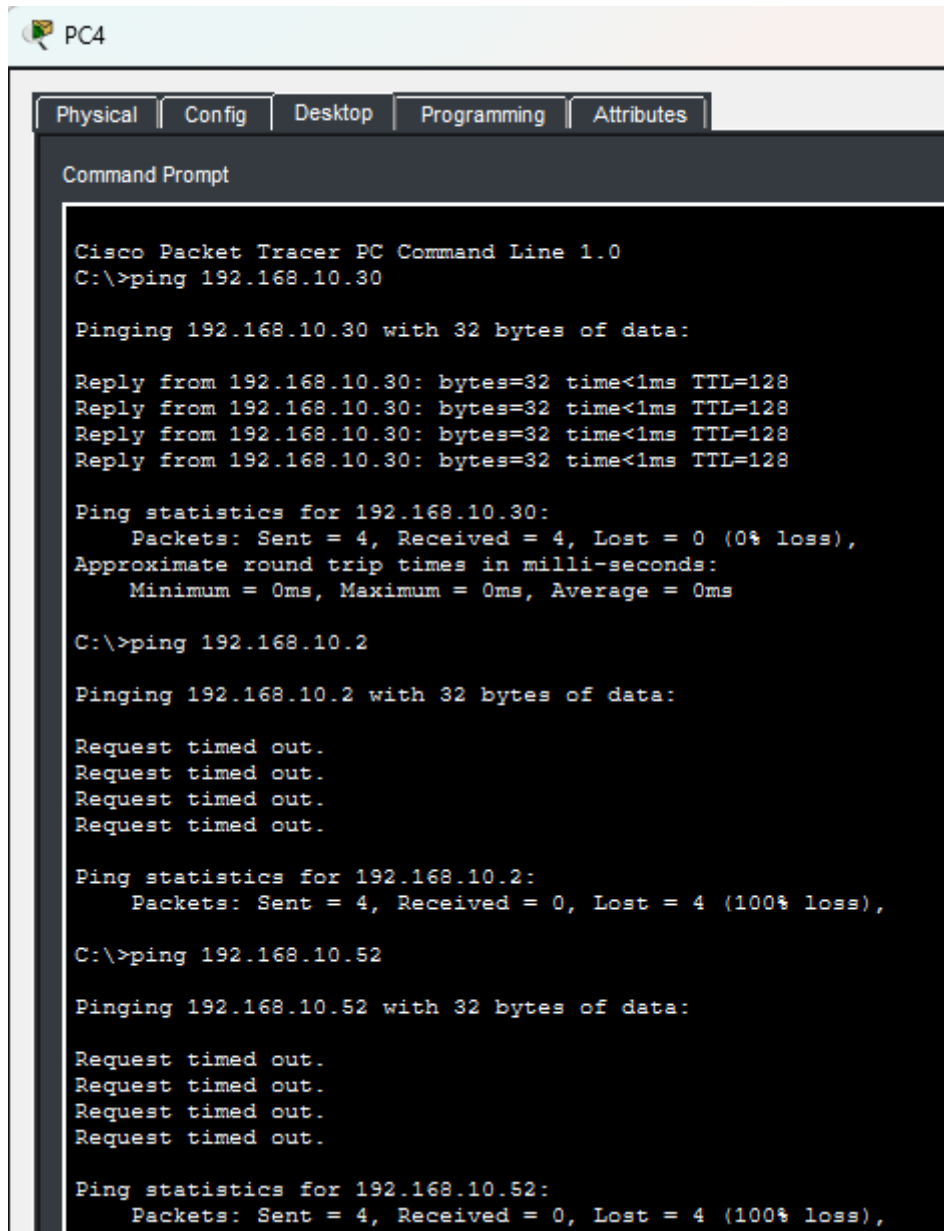
2-Desde PC4 (192.168.10.27) — Marketing (VLAN 3, SW1)

ping 192.168.10.30 - Marketing (OK, otro switch)

ping 192.168.10.2 - Ventas (FAIL)

ping 192.168.10.52 - RRHH (FAIL)

ping 192.168.10.28 – Marketing (OK)



```
PC4
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.30

Pinging 192.168.10.30 with 32 bytes of data:

Reply from 192.168.10.30: bytes=32 time<1ms TTL=128
Reply from 192.168.10.30: bytes=32 time<1ms TTL=128
Reply from 192.168.10.30: bytes=32 time<1ms TTL=128
Reply from 192.168.10.30: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

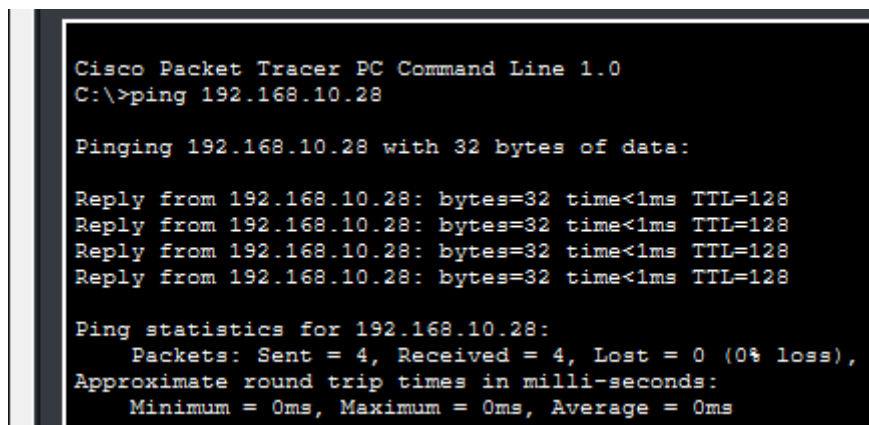
Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.52

Pinging 192.168.10.52 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.52:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.28

Pinging 192.168.10.28 with 32 bytes of data:

Reply from 192.168.10.28: bytes=32 time<1ms TTL=128
Reply from 192.168.10.28: bytes=32 time<1ms TTL=128
Reply from 192.168.10.28: bytes=32 time<1ms TTL=128
Reply from 192.168.10.28: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.28:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

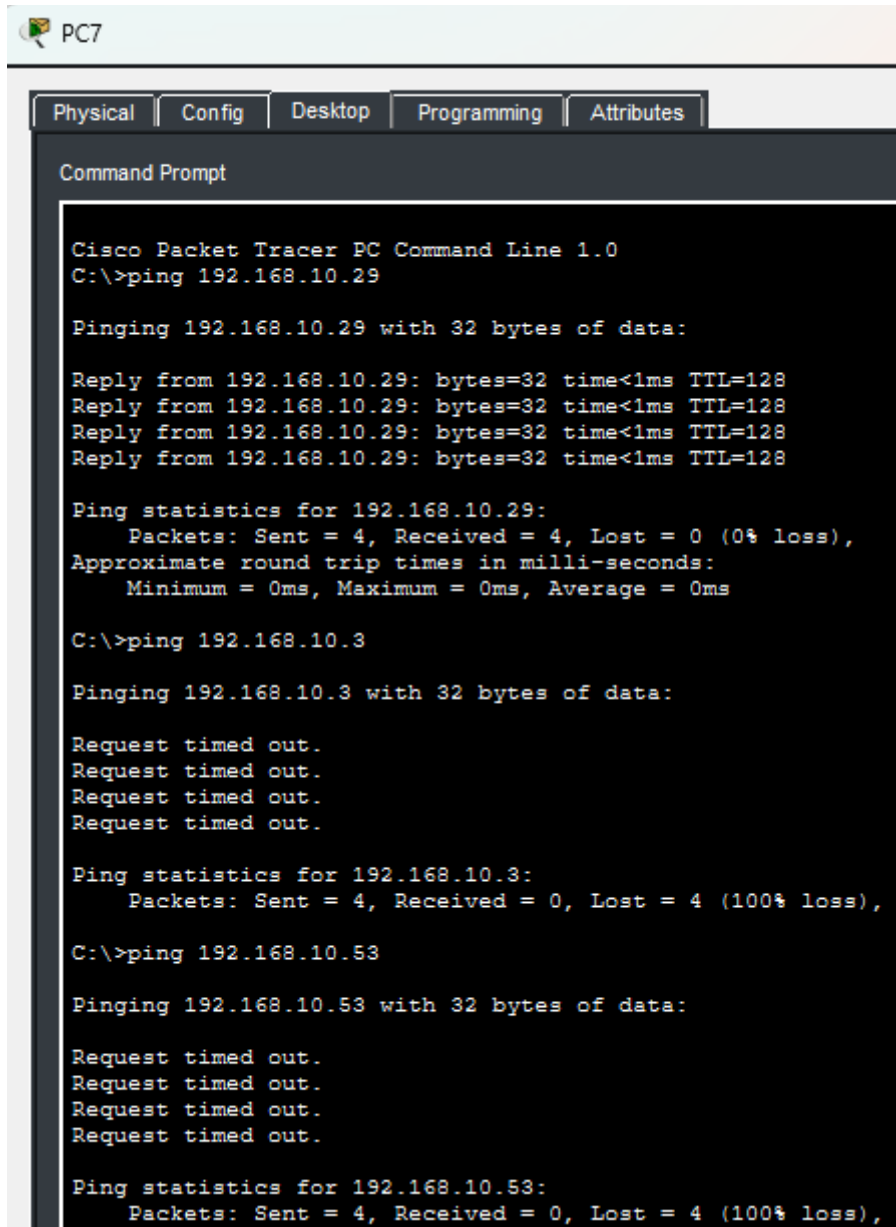
3-Desde PC7 (192.168.10.30) — Marketing (VLAN 3, SW2)

ping 192.168.10.29 - Marketing (OK)

ping 192.168.10.3 - Ventas (FAIL)

ping 192.168.10.53 - RRHH (FAIL)

ping 192.168.10.28 – Marketing (OK, otro switch)



```
PC7

Physical | Config | Desktop | Programming | Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.29

Pinging 192.168.10.29 with 32 bytes of data:

Reply from 192.168.10.29: bytes=32 time<1ms TTL=128
Reply from 192.168.10.29: bytes=32 time<1ms TTL=128
Reply from 192.168.10.29: bytes=32 time<1ms TTL=128
Reply from 192.168.10.29: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.29:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

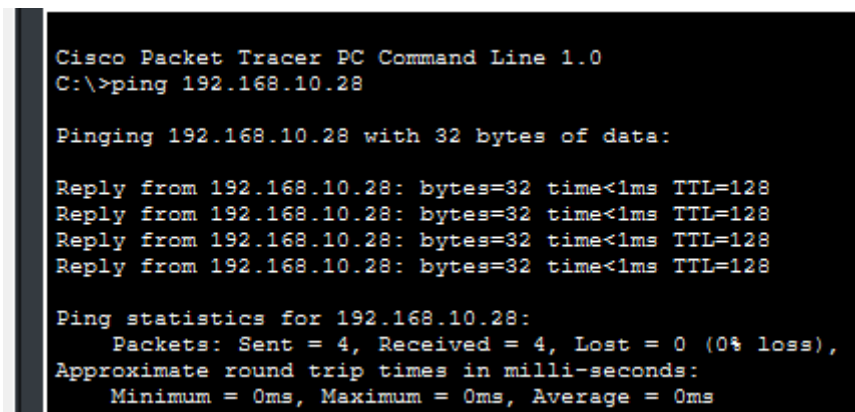
Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.53

Pinging 192.168.10.53 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.53:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.28

Pinging 192.168.10.28 with 32 bytes of data:

Reply from 192.168.10.28: bytes=32 time<1ms TTL=128
Reply from 192.168.10.28: bytes=32 time<1ms TTL=128
Reply from 192.168.10.28: bytes=32 time<1ms TTL=128
Reply from 192.168.10.28: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.28:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

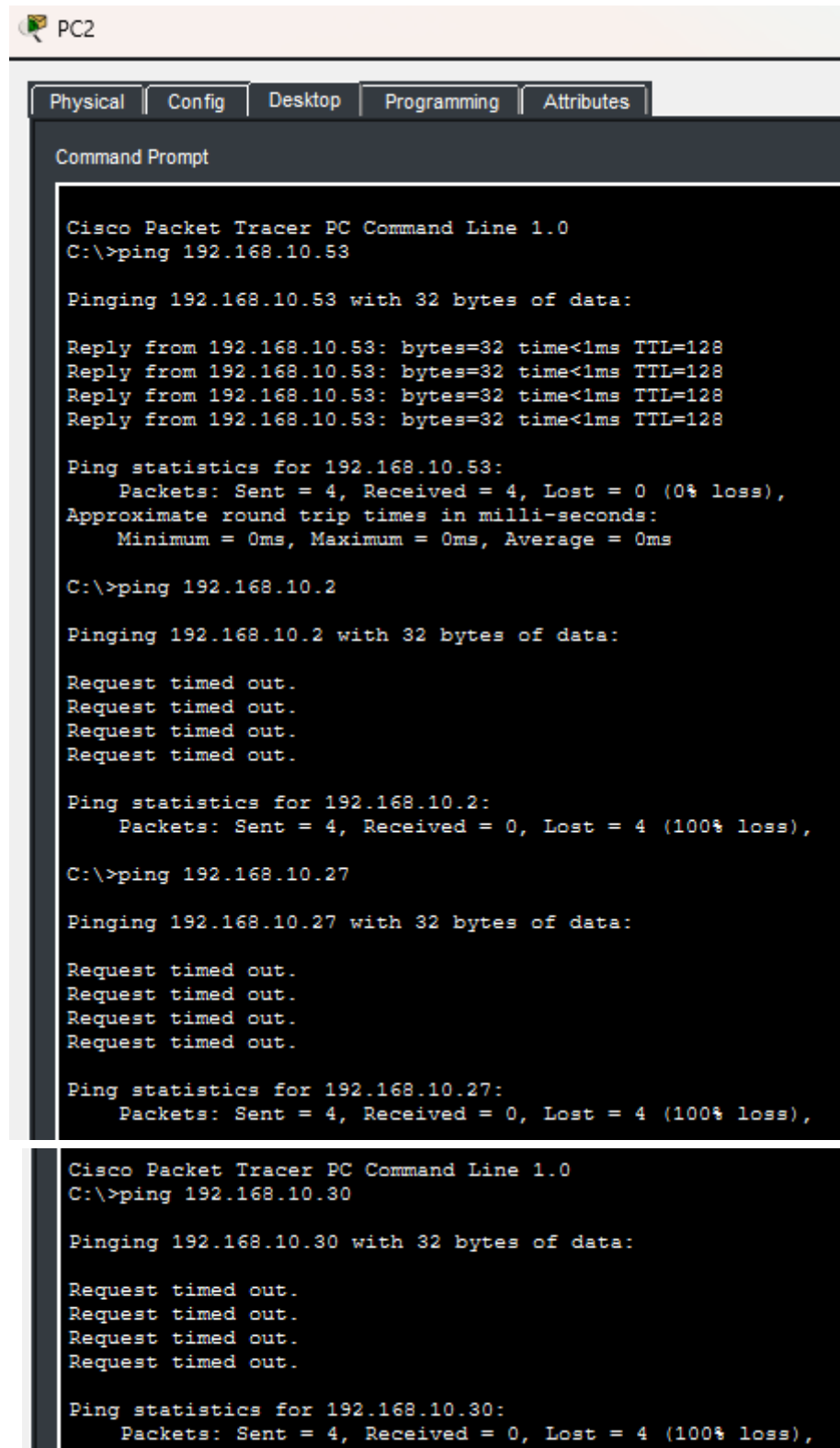

4-Desde PC2 (192.168.10.52) — RRHH (VLAN 4)

ping 192.168.10.53 - RRHH (OK)

ping 192.168.10.2 - Ventas (FAIL)

ping 192.168.10.27 - Marketing switch 1 (FAIL)

ping 192.168.10.30 - Marketing switch 2 (FAIL)



The screenshot shows the PC2 interface in Cisco Packet Tracer. The 'Config' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the results of four ping commands executed from PC2 (192.168.10.52).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.53

Pinging 192.168.10.53 with 32 bytes of data:

Reply from 192.168.10.53: bytes=32 time<1ms TTL=128
Reply from 192.168.10.53: bytes=32 time<1ms TTL=128
Reply from 192.168.10.53: bytes=32 time<1ms TTL=128
Reply from 192.168.10.53: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.53:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.27

Pinging 192.168.10.27 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.27:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.30

Pinging 192.168.10.30 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

7-Conclusiones

Con las pruebas de ping se comprueba que los equipos del mismo departamento sí se comunican y los de distintos no. Además, Marketing funciona entre los dos switches, lo que confirma que el enlace troncal está bien configurado y que la segmentación por VLAN funciona correctamente.

Autor: Luis Mateos

Curso: ASIR 1º

Asignatura: Redes